DRIDNet: Diffusion-Regularized Intrinsic Decomposition Network for Shadow Robust Hyperspectral and LiDAR Classification

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***Abstract*—Multimodal remote sensing data, especially hyperspectral image (HSI) and light detection and ranging (LiDAR) data, have substantially advanced land-cover classification. However, cloud shadows and structure-induced occlusions distort observed spectra and also weaken cross-modal consistency, which degrades classification performance in complex scenes. Most existing HSI–LiDAR classification methods learn directly from observed radiance by using convolutional or transformer backbones, while paying limited attention to the physical image-formation process and illumination-related variability. To address this issue, we propose a diffusion-regularized intrinsic decomposition network (DRIDNet) for shadow-robust HSI and LiDAR classification. At the physical level, a coordinate implicit intrinsic decomposition module (CIIDM) decomposes radiance into an intrinsic reflectance component and a shading-related environmental component, and a spectral denoising diffusion probabilistic model (DDPM) prior regularizes reflectance spectra toward a plausible material manifold, thereby suppressing noise-driven and nonphysical solutions in shadowed regions. At the representation level, a dual-stream pyramid encoder extracts multiscale spectral–spatial cues from HSI and structural cues from LiDAR, while a cross-modal transport routing module (CTRM) performs content-adaptive alignment and routing on high-level scale-aligned multimodal features. Experiments on four public HSI–LiDAR datasets demonstrate that DRIDNet consistently improves over representative baselines. In addition, a dedicated shadow-region evaluation on Houston2013 shows that the proposed method remains more robust than competing approaches, with overall accuracy (OA) gains of 0.65 percentage points on the full shadow region and 14.71 percentage points on a representative hard local shadow patch.**

***Index Terms*—Diffusion prior, hyperspectral imagery, intrinsic decomposition, land-cover classification, light detection and ranging (LiDAR), multimodal fusion, optimal transport.**

I. Introduction

REMOTE sensing technology has become a key noncontact technology for acquiring and analyzing information about the Earth's surface. With the rapid development of Earth observation satellites and airborne platforms, multimodal remote sensing data can now be collected synchronously over the same scene from multiple platforms, providing a rich information basis for fine land cover mapping [1], refined urban management [2], precision agriculture [3], and ecological monitoring [4]. Among these modalities, hyperspectral images (HSIs) offer dense and continuous spectral bands that capture subtle material differences in the spectral domain, whereas light detection and ranging (LiDAR) data record 3-D elevation and structural information that is crucial for describing complex object shapes and terrain relief. This complementarity between spectral and geometric information has made joint classification of HSI and LiDAR data an important direction for multisource remote sensing interpretation [5], [6].

Methodologically, joint HSI and LiDAR classification has evolved from handcrafted fusion toward unified multimodal representation learning. Early pipelines mainly relied on traditional machine learning and adopted feature-level fusion (FLF) and decision-level fusion (DLF) to integrate descriptors or decisions from each modality for improved robustness [7], [8]. With continued advances in deep learning, multimodal fusion methods have shifted toward learnable representations. Representative categories include CNN-based fusion networks that use multibranch architectures to integrate spatial and spectral modeling of HSI with elevation and structural cues from LiDAR, enabling automated feature extraction and fusion in a unified pipeline [9]. Under limited annotation, semi-supervised multimodal fusion methods exploit pseudo labels and representation learning to reduce overfitting and improve training stability under weak supervision [10], [11]. To capture global context and long-range dependencies, transformer-based fusion designs have been introduced to perform attention-based alignment between token representations of HSI and LiDAR [12], [13], [14]. In addition, contrastive learning and text-supervised training have been introduced to improve cross-dataset transferability and robustness to distribution shifts [15], [16]. Moreover, graph-based multimodal learning provides relational modeling on irregular spatial structures and supports neighborhood aggregation beyond regular grids, which is particularly relevant for remote sensing scenes [17], [18]. These advances have substantially improved multimodal representation capacity, yet most existing designs still treat the observed hyperspectral signal as the primary

Received 14 May 2026; revised 16 July 2026; accepted 11 August 2026. Date of publication 14 August 2026; date of current version 3 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62575028 and Grant U2241275. *(Corresponding author: Kun Gao.)*

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Digital Object Identifier 10.1109/TGRS.2026.3723811

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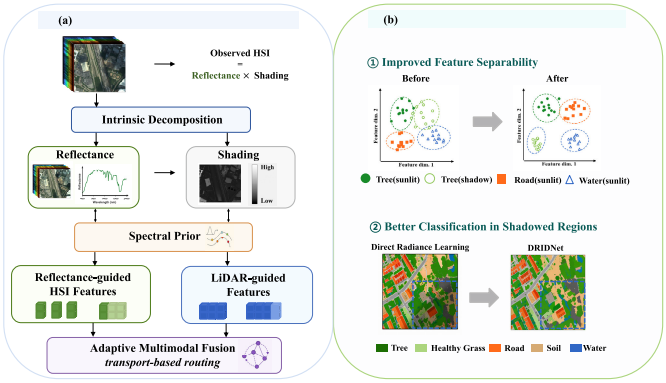


Fig. 1. Conceptual overview of DRIDNet for shadow-robust HSI–LiDAR classification. (a) DRIDNet decomposes the observed HSI into reflectance and shading, regularizes reflectance with a spectral prior, and extracts reflectance-guided and LiDAR-guided features for adaptive multimodal fusion with transport-based routing. (b) By reducing shadow-induced feature confusion, DRIDNet improves feature separability and classification reliability in shadowed regions.

learning target and rely on data-centric normalization to handle environmental variability.

Despite substantial progress, hyperspectral observations in practical remote sensing are inevitably affected by environmental factors, including atmospheric propagation, cloud occlusion, and shadowing effects caused by scene geometry. These factors modulate the measured radiance before feature extraction and make the observed HSI deviate from the material-related reflectance that is more directly associated with land-cover categories. Therefore, shadow-robust HSI–LiDAR classification requires not only stronger cross-modal fusion, but also explicit consideration of how illumination- and geometry-related variations enter the hyperspectral signal. Several common limitations remain under complex scenes.

1) Many methods learn directly from observed radiance or normalized HSI features, without explicitly distinguishing material-related spectral information from illumination, topography, and other environmental factors. This makes the learned representation sensitive to shadow occlusion and acquisition variations.
2) Most fusion architectures perform cross-modal interaction on observed HSI and LiDAR features through concatenation, attention, or fixed fusion rules. However, the reliability of spectral and geometric cues varies across land-cover classes, spatial regions, and shadow conditions, requiring more adaptive modality allocation.
3) Existing approaches remain largely data-driven and seldom constrain spectral representations with physically meaningful priors. As a result, shadow-induced variations or noise-driven spectral patterns may be fit by the model, limiting physical consistency and statistical robustness under limited supervision and cross-scene shifts.

As illustrated in Fig. 1, the observed HSI can be understood as a coupled result of intrinsic reflectance and shading-related environmental effects. Under shadowed conditions, radiance-level features may mix material information with illumination variation, thereby enlarging within-class spectral differences and weakening the separability of land-cover categories. Since shading is closely related to scene geometry and terrain structure, LiDAR provides complementary cues for interpreting these environment-induced variations. This motivates a physically grounded classification strategy that goes beyond feature-level fusion alone, where reflectance recovery, spectral plausibility, and geometry-aware multimodal interaction are jointly considered before final classification.

Along this direction, hyperspectral intrinsic image decomposition (HIID), together with physics-based and statistics-based priors, has emerged as a complementary line of research that enhances robustness by separating reflectance from environmental effects and producing representations closer to intrinsic material properties [19], [20]. The core idea extends intrinsic image theory to hyperspectral observations by decomposing radiance into reflectance and an environmental component that includes shading among other factors [21], and learning-based formulations further improve the consistency of the decomposition in complex environments [22]. Early intrinsic decomposition in this vein can be traced to nonlocal, prior-driven formulations such as NonLocalID [23], which motivates subsequent physics- and geometry-aware extensions. For multimodal tasks, researchers have explored intrinsic decomposition-guided fusion to reduce modality-induced distribution shifts by coupling intrinsic components with LiDAR geometry, as demonstrated by HI2D2FNet [24]. However, existing methods mainly focus on decomposition consistency or geometry-aware intrinsic fusion, while the classification-oriented use of recovered reflectance under shadows remains insufficiently explored.

Meanwhile, diffusion-inspired modeling provides a generative way to regularize spectra toward physically plausible manifolds [25], [26], [27]. Compared with hand-crafted smoothness constraints, diffusion-based priors can capture global interband correlations and material-dependent spectral patterns. These advances motivate jointly considering reflectance recovery, spectral prior regularization, and adaptive HSI–LiDAR fusion for shadow-robust classification.

Based on the above analysis, this work presents a joint HSI and LiDAR classification framework, which yields physically grounded and statistically robust representations by explicitly separating reflectance from environmental effects and regularizing spectra with a diffusion-based prior. The fusion stage performs data-dependent, interpretable conditional computation that allocates modality importance across classes and regions. Therefore, the diffusion-regularized intrinsic decomposition network (DRIDNet) improves classification accuracy and stability in the presence of cloud shadow occlusion and shadowed regions on the ground surface.

The main contributions of this article are summarized as follows.

1) We propose DRIDNet, a unified framework for joint HSI and LiDAR classification, which improves classification accuracy and stability under cloud shadows and terrain-induced illumination variations.
2) We develop a coordinate implicit intrinsic decomposition module (CIIDM) to separate material reflectance from

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a shading-related environmental component by leveraging LiDAR geometry cues before downstream feature learning.

3) We introduce a spectral denoising diffusion probabilistic model (DDPM) prior, referred to as the spectral DDPM prior (SDDP), as a training-time regularization strategy to guide the recovered reflectance toward a physically plausible spectral manifold and improve robustness under limited supervision and distribution shifts.
4) We design a dual-stream pyramid feature encoding module (DPFEM) and a cross-modal transport routing module (CTRM) for reflectance-conditioned HSI–LiDAR fusion, enabling multiscale spectral–geometric representation and balanced modality allocation across classes and regions.

The remainder of this article is organized as follows. Section II reviews related work on joint HSI–LiDAR classification, HIID, and diffusion-based spectral priors. Section III presents the overall architecture of the proposed method. Section IV describes the experimental setup, results, and ablation studies. Section V concludes this article and discusses limitations and future directions.

II. Related Work

A. Joint HSI and LiDAR Classification Methods

Early FLF and DLF pipelines provide the conceptual basis for multimodal fusion by separately constructing modality-specific descriptors and then combining them in the feature space or the decision space [7], [8]. With deep learning, representative multimodal fusion architectures have been developed across CNNs, transformers, and graphs. Wang et al. [28] proposed S3F2Net with coordinated branches for spatial context, spectral signatures, and structural geometry, while graph convolution is incorporated to enhance cross-modal consistency. Ni et al. [29] designed FDNet by mapping HSI patches and LiDAR patches into the frequency domain to learn complementary frequency components. Tang et al. [9] developed MICF-Net to jointly exploit spatial and spectral cues from HSI and elevation information from LiDAR, and to integrate the two modalities through FLF and DLF. Graph-based methods further model structural relationships beyond regular grids, including RVFL with graph convolutional embeddings [30], feature-guided dynamic graph construction [31], and graph attention with local spectral filtering [32].

For robustness under limited supervision, multiscale fusion refinement, contrastive alignment, and pseudolabel-guided semi-supervised learning have been explored [33], [34], [35]. To enhance global and long-range modeling, Zhao et al. [12] proposed BIHAF-Net with hierarchical self-attention and cross-modal attention alignment, while Zhang et al. [36] developed an efficient Mamba-based network for HSI–LiDAR joint classification. Recent work has also revisited frequency-domain complementarity, with Liu et al. [37] introducing progressive spatial–spectral–frequency fusion to integrate spatial, spectral, and frequency representations.

Recent multisensor remote sensing studies also suggest that robust multimodal analysis benefits from considering imaging mechanisms and degradation factors [38], [39]. Although these studies address different tasks, they highlight a limitation of many existing HSI–LiDAR classification networks: cross-modal feature interaction is strengthened, whereas illumination- and shadow-induced degradation in the observed hyperspectral radiance is still not explicitly modeled before fusion.

B. HIID and Physical Priors

HIID extends intrinsic image theory to hyperspectral observations by decomposing radiance into reflectance and the environmental component, including shading, thereby yielding physically consistent representations for downstream tasks [19].

Representative HIID methods include physics-guided and geometry-aware designs. Xie et al. [40] developed a physics-guided unsupervised HIID model that explicitly embeds imaging models and constraint terms to improve stability. Jin et al. [41] introduced DSM into intrinsic decomposition through IHID with DSM to enforce terrain-consistent intrinsic components, and Huang et al. [42] developed DSM-HIIDU to jointly perform intrinsic decomposition and unmixing under DSM-based energy constraints. Gu et al. [43] proposed PAN-HIID by incorporating high-resolution panchromatic imagery to enhance spatial detail in reflectance while preserving spectral fidelity. For discriminative learning, Gong et al. [44] developed AdverDecom and subsequently introduced HyperDID to enlarge interclass differences and reduce within-class variability under challenging illumination [45].

For HSI and LiDAR joint classification, intrinsic decomposition and priors have been integrated into multisource fusion in several forms. Yu et al. [46] developed SMD-MIID to align LiDAR geometry with shading in the intrinsic domain using a shadow mask, and subsequently proposed PID-HLfusion to exploit modality-specific illumination and connect hyperspectral shading geometry with LiDAR elevation [47]. Yu and Zhang [48] proposed a cross-modal 3-D intrinsic framework to jointly represent reflectance, shading, and geometric structure. In parallel, diffusion model-based hyperspectral super-resolution and fusion approaches construct generative processes in the hyperspectral domain [49]. These methods provide statistical priors that reconstruct fine-grained reflectance under physical constraints and facilitate alignment with LiDAR geometry.

These studies demonstrate the value of geometry-aware physical modeling for reducing illumination-induced variability. However, most of them focus on decomposition consistency or geometry-guided intrinsic fusion. For shadow-robust classification, it remains important to regularize recovered reflectance in the spectral domain and adaptively combine it with LiDAR geometry.

III. Methodology

A. Overall Framework of the Proposed Method

DRIDNet addresses joint HSI–LiDAR classification by combining physically informed decomposition, spectral prior regularization, and adaptive multimodal routing in a single

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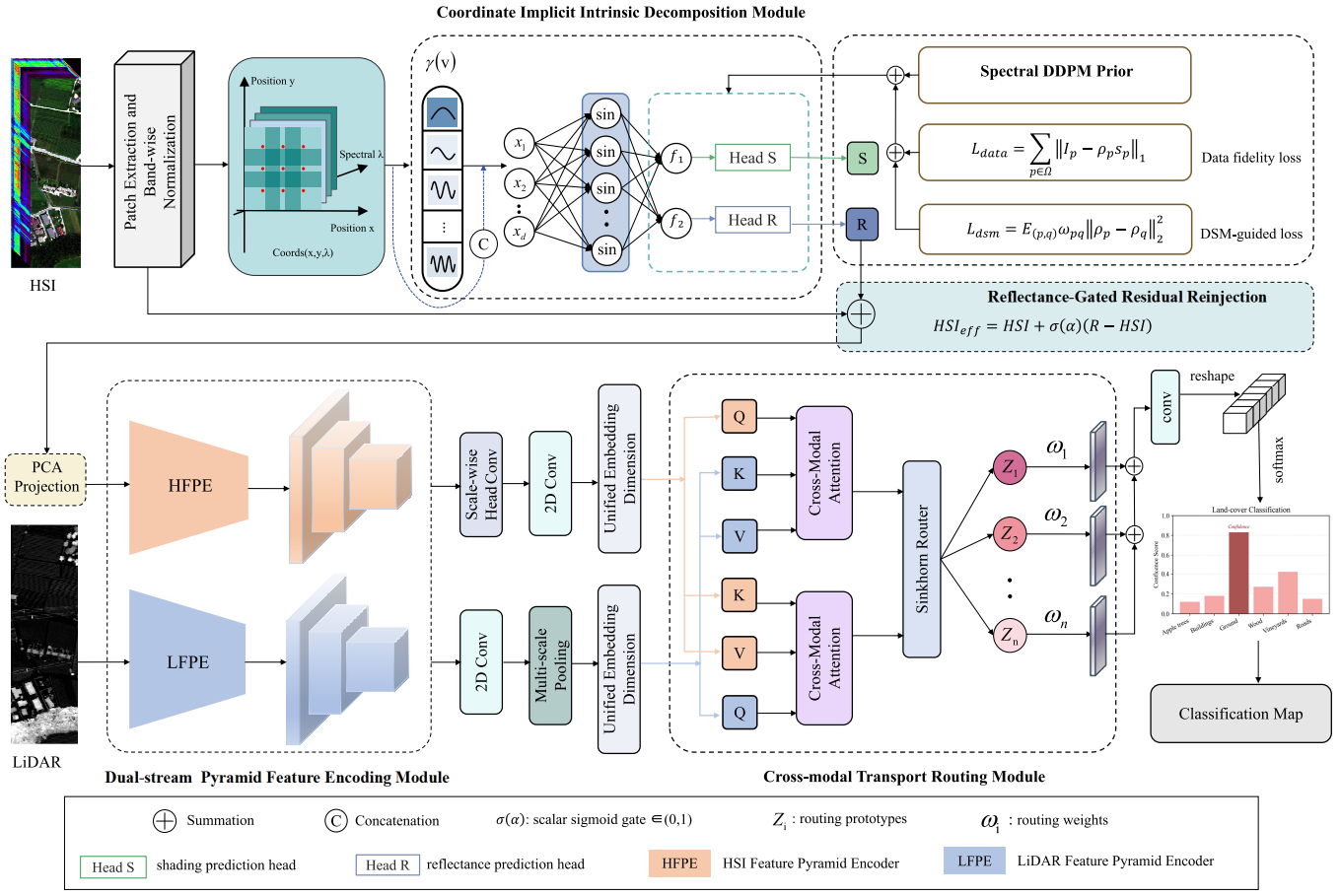


Fig. 2. Overall architecture of the proposed framework.

end-to-end framework. As shown in Fig. 2, the proposed method contains four main components: the CIIDM, the SDDP, the DPFEM, and the CTRM.

Given a spatially registered HSI $\mathbf{Y} \in \mathbb{R}^{H\times W\times B}$ and a corresponding LiDAR image $\mathbf{D} \in \mathbb{R}^{H\times W\times 1}$, we extract paired HSI and LiDAR patches of size $r \times r$, denoted by $\mathbf{p}_h \in \mathbb{R}^{r\times r\times B}$ and $\mathbf{p}_l \in \mathbb{R}^{r\times r\times 1}$. The HSI patch is first processed by CIIDM in the original sensor-band space. CIIDM estimates a reflectance component and a shading-related environmental component under a log-domain image-formation model. The recovered reflectance is then reinjected into the HSI branch through a learnable gated residual connection, producing an effective hyperspectral representation HSI_{eff}. This representation is not a pure reflectance spectrum; instead, it adaptively combines the original observed HSI and the CIIDM-recovered reflectance, thereby preserving original spectral evidence while suppressing illumination-induced distortions. After this reinjection step, PCA is applied to HSI_{eff} to obtain a compact HSI representation for subsequent feature encoding. Therefore, PCA is used as a post-CIIDM spectral compression step rather than as a substitute for physical reflectance modeling.

During training, the recovered reflectance is additionally constrained by SDDP. Specifically, SDDP is a 1-D diffusion model trained on library-derived reflectance vectors and used only as a training-time prior. The library spectra are first convolved with the sensor spectral responses to obtain reflectance vectors in the same B-band sensor space as the input HSI. Therefore, SDDP learns not only local smoothness but also global interband dependencies, characteristic absorption structures, and material-dependent spectral distributions. It regularizes per-pixel reflectance spectra in the original B-band domain before PCA compression, thereby improving the physical plausibility and statistical stability of decomposition outputs without introducing reverse-sampling cost at inference time.

The resulting PCA-projected HSI patch and the paired LiDAR patch are then fed into DPFEM. The HSI stream uses 3-D convolutions to preserve spectral structure within the reduced subspace, whereas the LiDAR stream uses 2-D convolutions to emphasize structural and elevation cues. Both streams construct multiscale feature pyramids and project them to a unified embedding dimension.

Finally, the top-level scale-aligned HSI and LiDAR features produced by DPFEM are rearranged into token sequences and fed into CTRM for cross-modal interaction and routing. A linear cross-modal attention block first calibrates the top-level HSI and LiDAR embeddings through local bidirectional spectral–geometric interaction. Since cross-attention itself does not impose a global allocation constraint on the fused tokens, the calibrated features are further routed by

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an entropy-regularized transport module with learnable expert prototypes. The OT marginal constraints encourage balanced utilization of fusion experts and reduce over-concentrated routing in shadowed or structurally complex regions. The aggregated routed features are sent to a lightweight classification head, which produces the final class logits after spatial pooling. In this way, DRIDNet explicitly separates physical disentanglement, spectral regularization, multiscale encoding, and high-level adaptive cross-modal decision-making.

B. Coordinate Implicit Intrinsic Decomposition Module

To enable robust cross-modal fusion and subsequent routing, we first decouple the representation at the level of the physical image formation model to suppress confounding effects induced by illumination variation and terrain geometry. CIIDM adopts a continuous formulation over spatial and spectral coordinates and decomposes the observed radiance into material reflectance and shading under minimal assumptions and regularization. This design improves stability under cross-scene distribution shift and limited training samples.

We consider paired inputs $\mathbf{p}_h \in \mathbb{R}^{r\times r\times B}$ denoting an HSI patch in the original sensor-band domain with B bands and $\mathbf{p}_l \in \mathbb{R}^{r\times r\times 1}$ denoting the registered LiDAR cue. The observation model at each pixel-band location (x, y, λ) is formulated as

$$\mathbf{Y}(x, y, \lambda) = R(x, y, \lambda)\ S(x, y, \lambda) \tag{1}$$

where $R, S \in \mathbb{R}^{r\times r\times B}$ denote reflectance and shading, respectively. CIIDM represents R and S as continuous functions of normalized coordinates using a sinusoidal coordinate encoder and a lightweight implicit network. The implicit network is shared by all patches and is trained end-to-end with the classification framework, instead of being re-optimized for each individual patch. To avoid making the decomposition depend on coordinates alone, the reflectance branch is conditioned on HSI content features distilled from p_h. In parallel, LiDAR geometry is incorporated by conditioning the shading branch on features distilled from p_l, which encourages S to vary smoothly within regions while aligning transitions with terrain-induced boundaries. Output activations enforce value ranges without extra penalties, using sigmoid for $R \in [0, 1]$ and softplus for $S \geq 0$. For efficiency, coordinates are uniformly sampled or randomly sampled within each patch during training.

Physical consistency is imposed by a data fidelity term and a DSM-guided edge-aware smoothness term

$$\mathcal{L}_{\text{data}} = \sum_{p\in\Omega} \|\mathbf{Y}_p - R_p \odot S_p\|_1 \tag{2}$$

$$\mathcal{L}_{\text{dsm}} = \sum_{(p,q)\in\mathcal{E}} \omega_{pq} \|R_p - R_q\|_2^2 \tag{3}$$

where Ω denotes the pixel set within a patch, and $\mathcal{E}$ denotes a set of sampled adjacent pixel pairs. For $(p, q) \in \mathcal{E}$, the weight ω_{pq} is derived from DSM slope magnitude together with spectral and appearance contrast, so that strong boundaries receive smaller weights while homogeneous regions receive larger weights. The total CIIDM objective is

$$\mathcal{L}_{\text{CIIDM}} = \mathcal{L}_{\text{data}} + \lambda_{\text{dsm}}\, \mathcal{L}_{\text{dsm}} \tag{4}$$

where λ_{dsm} is tuned on a held-out validation set.

To balance physical consistency and discriminative power in downstream tasks, CIIDM applies a reflectance-gated residual reinjection into the hyperspectral branch. Let $\text{HSI} \in \mathbb{R}^{r\times r\times B}$ be the input radiance patch, and let $R \in \mathbb{R}^{r\times r\times B}$ be the recovered reflectance. We form an effective hyperspectral representation by a gated residual injection

$$\text{HSI}_{\text{eff}} = \text{HSI} + \sigma(\alpha)(R - \text{HSI}) \tag{5}$$

where $\alpha \in \mathbb{R}$ is a scalar gate coefficient and $\sigma(\alpha) \in (0, 1)$. As $\sigma(\alpha)$ approaches one, the pipeline relies more on R and suppresses illumination- and terrain-induced artifacts. As $\sigma(\alpha)$ approaches zero, the original observation is preserved to reduce error propagation from imperfect decomposition. The tensor HSI_{eff} preserves the original spectral layout of the input HSI. After the reinjection step, PCA is applied to HSI_{eff} to obtain a reduced hyperspectral representation $\text{HSI}_{\text{pca}} \in \mathbb{R}^{r\times r\times k}$ for downstream feature encoding, where k denotes the retained spectral dimension.

Overall, CIIDM performs explicit disentanglement at the level of the image formation model with minimal regularization and an adaptive reinjection mechanism, yielding a stable reflectance representation, a terrain-consistent shading field, and an HSI_{eff} that is further projected to HSI_{pca} for downstream cross-modal alignment and routing.

C. Spectral DDPM Prior

Intrinsic decomposition is statistically ill-posed. Under limited samples or distribution shift, data fidelity alone may yield nonphysical reflectance with oscillations, spurious absorption features, or inconsistent interband correlations. To mitigate this risk without increasing inference overhead, we introduce the SDDP, abbreviated as SDDP, as a training-time regularizer in reflectance space, as shown in Fig. 3. The prior is defined on the original spectral domain and constrains per-pixel reflectance vectors in $\mathbb{R}^B$. Importantly, this prior is imposed before any downstream PCA-based compression, so that reflectance regularization is performed in the original sensor-band space. Different from a hand-crafted smoothness loss, SDDP learns a data-driven distribution of plausible material reflectance curves from sensor-response-convolved spectral library samples. This distribution captures global interband correlations, material-dependent absorption patterns, and multimodal spectral variability, which are difficult to encode using only local derivative penalties or pairwise spectral consistency constraints.

To map a high-resolution library spectrum $r(\lambda)$ to the B sensor bands, we compute the relative spectral-response-weighted mean reflectance for each band i as

$$\bar{r}_i = \frac{\int r(\lambda)\, S_i(\lambda)\, \mathrm{d}\lambda}{\int S_i(\lambda)\, \mathrm{d}\lambda}, \qquad i = 1, \ldots, B \tag{6}$$

where $S_i(\lambda)$ denotes the relative spectral response of the ith sensor band, and B is the number of HSI bands of the corresponding dataset. The obtained $\bar{r}_i$ is clipped to [0, 1] to match the output range of the CIIDM reflectance branch. In this way, each high-resolution library spectrum is converted into a B-dimensional reflectance vector with the same band definition as the input HSI. Since the four datasets contain

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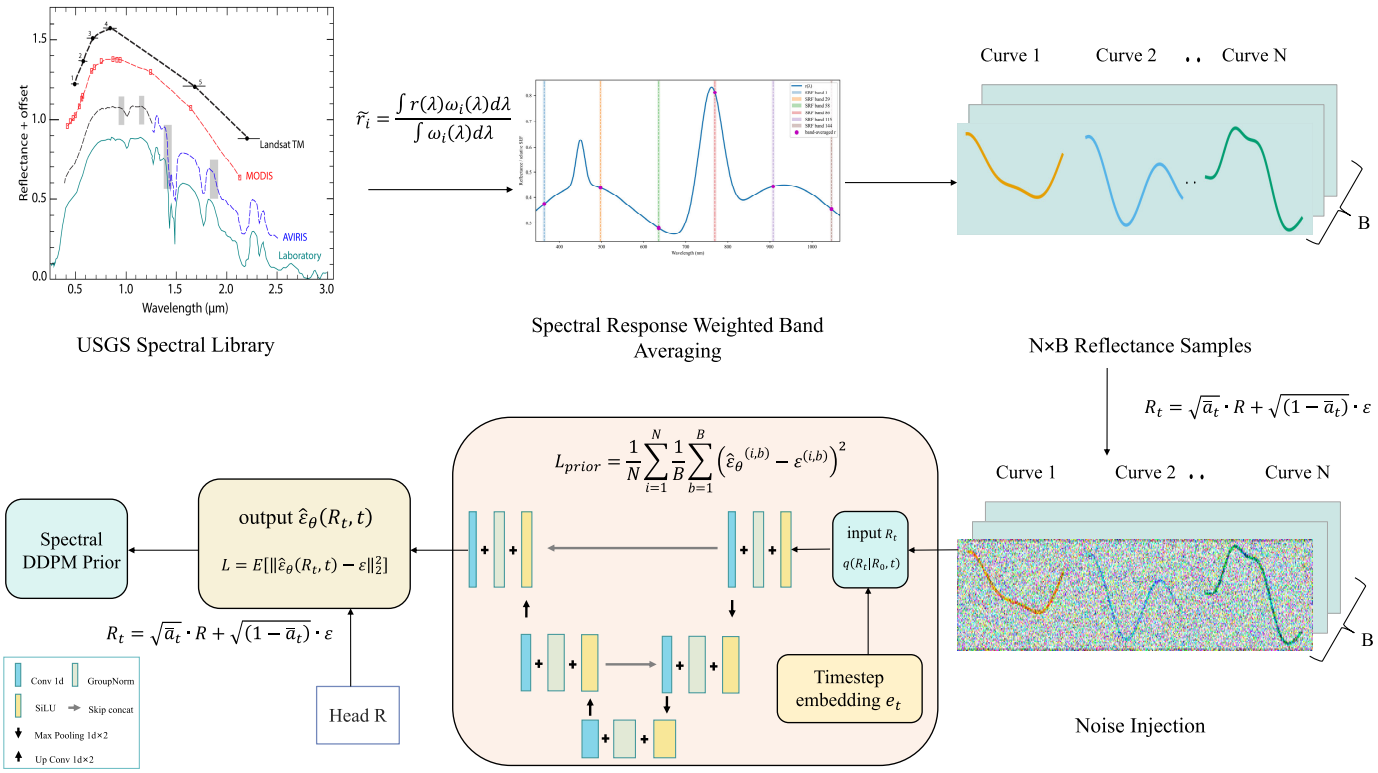


Fig. 3. Training-time SDDP for regularizing reflectance in the original B-band domain before downstream PCA-based compression.

different numbers of retained spectral bands, SDDP is instantiated separately in the corresponding sensor-band space of each dataset, with dimensionalities of 144, 48, 64, and 63 for Houston2013, Houston2018, MUUFL Gulfport, and Trento, respectively.

SDDP is a DDPM trained on B-dimensional reflectance vectors. The forward diffusion process is

$$q(R_t \mid R) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, R,\ (1-\bar{\alpha}_t)\,\mathbf{I}\right) \tag{7}$$

and the training objective is

$$\mathcal{L}_{\text{prior}} = \mathbb{E}_{t,R,\varepsilon}\left[\|\varepsilon - \varepsilon_\theta(R_t, t)\|_2^2\right] \tag{8}$$

where ε_θ is a 1-D U-Net noise predictor.

We train SDDP on library-derived reflectance vectors and then freeze it during joint training with CIIDM. During joint training, the CIIDM reflectance spectrum $\hat{\mathbf{r}} \in \mathbb{R}^B$ at each pixel is treated as a clean sample. A timestep t and noise ε are drawn to construct

$$R_t = \sqrt{\bar{\alpha}_t}\,\hat{\mathbf{r}} + \sqrt{1-\bar{\alpha}_t}\,\varepsilon \tag{9}$$

after which $\mathcal{L}_{\text{prior}}(\hat{\mathbf{r}})$ is backpropagated jointly with $\mathcal{L}_{\text{CIIDM}}$. Minibatches contain N reflectance spectra of length B arranged as an $N \times B$ matrix. At inference, we do not run the reverse sampler. SDDP is used solely as a training-time regularizer on reflectance vectors, thereby introducing no additional inference cost.

By shaping a manifold of physically plausible reflectance curves with smooth variation, stable absorption features, and consistent interband correlations in the original sensor-band reflectance space, SDDP suppresses pathological solutions and improves robustness under small-sample and cross-scene conditions.

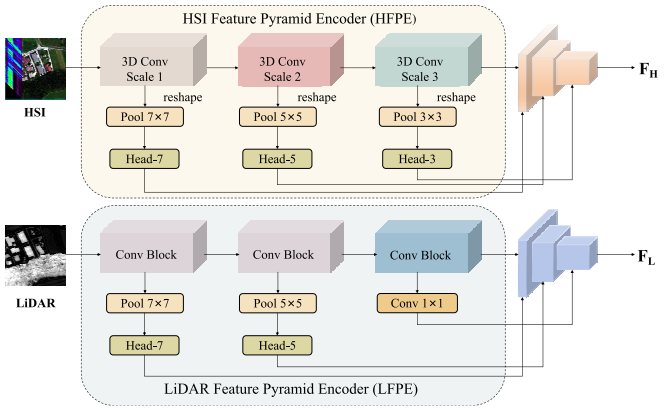


Fig. 4. Overview of the DPFEM.

D. Dual-Stream Pyramid Feature Encoding Module

Urban land cover classification relies on two complementary sources of evidence, including pixel-wise spectral signatures from hyperspectral imagery and neighborhood-level geometric context from LiDAR. Single-stream encoders and unweighted feature concatenation often sacrifice either spectral discriminability or geometric stability, while mismatched spatial scales hinder cross-modal fusion. To address these issues, we adopt a dual-stream pyramid encoder, as shown in Fig. 4. The HSI feature pyramid encoder and the LiDAR feature pyramid encoder embed modality-specific inductive biases and output three scale-aligned feature maps with a unified channel width, which serve as standardized inputs for subsequent cross-modal routing.

Given a PCA-projected effective hyperspectral patch $\text{HSI}_{\text{pca}} \in \mathbb{R}^{r\times r\times k}$ obtained after reflectance-gated residual

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reinjection and a co-registered LiDAR patch $\mathbf{p}_l \in \mathbb{R}^{r\times r\times 1}$, the dual-stream encoder builds hierarchical spectral–spatial and geometric pyramids with progressive spatial downsampling. The HSI stream uses 3-D convolutions with unit stride along the spectral axis to preserve spectral structure within the reduced subspace, whereas the LiDAR stream uses 2-D convolutions to emphasize long-range structure, slope transitions, and salient boundaries. Let $U_C(\cdot)$ be the channel width to C, and let $D_s^{3D}(\cdot)$ and $D_s^{2D}(\cdot)$ be downsampling operators at scale s for the HSI and LiDAR streams, respectively. The backbone features at scale $s \in \{1, 2, 3\}$ are defined as

$$\mathbf{F}_H^{(s)} = U_C\left(D_s^{3D}\left(\mathrm{HSI}_{\mathrm{pca}}\right)\right) \tag{10}$$

$$\mathbf{F}_L^{(s)} = U_C\left(D_s^{2D}\left(\mathbf{p}_l\right)\right). \tag{11}$$

To inject multiscale scene context consistently across streams, we aggregate structured pooled context at each stage and reinject it into the backbone features of each modality. For each scale s and index m, define $\mathcal{P}_{s,m}(\cdot) = U(\phi_{s,m}(\mathrm{Pool}_{s,m}(\cdot)))$, where $\mathrm{Pool}_{s,m}(\cdot)$ is structured pooling, $\phi_{s,m}$ is a linear mapping, and $U(\cdot)$ upsamples to the backbone resolution. The aggregated context is reinjected in a residual form

$$\bar{\mathbf{F}}_b^{(s)} = \mathrm{Norm}\left(\mathbf{F}_b^{(s)} + \sum_m \alpha_{s,m}\,\mathcal{P}_{s,m}\left(\mathbf{F}_b^{(s)}\right)\right) \tag{12}$$

where $b \in \{H, L\}$ denotes the modality. The weights $\alpha_{s,m}$ are nonnegative and normalized across m, which can be implemented by a softmax over learnable logits. This yields an HSI pyramid that maintains spectral discriminability while promoting region-consistent semantics, and an LiDAR pyramid that remains sensitive to geometric boundaries while attenuating spurious textures.

Before cross-modal interaction, both branches are projected to a common channel width C at each scale. With scale-wise projections $P_H^{(s)}$ and $P_L^{(s)}$, the aligned features are

$$\mathbf{Z}_H^{(s)} = \mathrm{Norm}\left(P_H^{(s)}\,\bar{\mathbf{F}}_H^{(s)}\right) \tag{13}$$

$$\mathbf{Z}_L^{(s)} = \mathrm{Norm}\left(P_L^{(s)}\,\bar{\mathbf{F}}_L^{(s)}\right). \tag{14}$$

Finally, the aligned tensors at the top pyramid level are rearranged into paired token sequences with matched lengths and dimensionalities, which are then fed into CTRM for cross-modal fusion and routing.

E. Cross-Modal Transport Routing Module

Building on the top-level scale-aligned features produced by DPFEM, we employ the CTRM, abbreviated as CTRM, to perform end-to-end cross-modal alignment and content-adaptive conditional computation under controlled parameter and computational budgets, ultimately producing per-pixel class probabilities. We consider hyperspectral and LiDAR token sequences $\mathbf{H}, \mathbf{L} \in \mathbb{R}^{N\times C}$ from the top pyramid level, where the hyperspectral tokens are derived by DPFEM from the PCA-projected effective hyperspectral representation $\mathrm{HSI}_{\mathrm{pca}}$, N is the number of tokens at that scale, and C is the unified channel width. Attention is parameterized by projection matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{C\times d}$. We compute bidirectional cross-modal attention to align spectral semantics with geometric cues at corresponding spatial locations

$$\widetilde{\mathbf{H}} = \mathrm{softmax}\left(\frac{(\mathbf{H}\mathbf{W}_Q)(\mathbf{L}\mathbf{W}_K)^\top}{\sqrt{d}}\right)(\mathbf{L}\mathbf{W}_V) \tag{15}$$

$$\widetilde{\mathbf{L}} = \mathrm{softmax}\left(\frac{(\mathbf{L}\mathbf{W}_Q)(\mathbf{H}\mathbf{W}_K)^\top}{\sqrt{d}}\right)(\mathbf{H}\mathbf{W}_V). \tag{16}$$

We then concatenate the updated tokens and apply a linear projection to obtain a calibrated representation

$$\mathbf{X} = \phi\left(\mathrm{Concat}\left(\widetilde{\mathbf{H}}, \widetilde{\mathbf{L}}\right)\right) \tag{17}$$

where $\phi(\cdot)$ is a learnable linear mapping. This step reduces inductive bias mismatch by enabling hyperspectral tokens to incorporate geometric constraints on boundaries and shapes, while allowing geometric tokens to exploit material-specific spectral evidence carried by the PCA-projected effective hyperspectral representation.

However, cross-attention mainly performs unconstrained pairwise interaction between HSI and LiDAR tokens, and does not explicitly regularize how the fused tokens are globally assigned to different discriminative fusion patterns. This may lead to over-concentrated routing, especially in shadowed regions or structurally complex areas where spectral and geometric cues are unevenly reliable.

Therefore, to achieve differentiable and balanced conditional computation, we introduce entropy-regularized optimal transport routing on $\mathbf{X} \in \mathbb{R}^{N\times C}$. In this formulation, the calibrated multimodal tokens are regarded as transported mass, the learnable expert prototypes are regarded as receiving bins, and the transport cost measures their spectral–geometric feature discrepancy. The marginal constraints introduce a global balance prior, which encourages stable and noncollapsed allocation of fused HSI–LiDAR information. The module maintains n learnable expert prototypes $\{\mathbf{Z}_k\}_{k=1}^n \subset \mathbb{R}^C$. The cost matrix $\mathbf{C} \in \mathbb{R}^{N\times n}$ measures the discrepancy between tokens and prototypes using squared Euclidean distance or cosine dissimilarity. We compute the soft assignment matrix $\mathbf{\Pi} \in \mathbb{R}_+^{N\times n}$ by solving the entropy-regularized optimal transport problem with uniform marginals

$$\min_{\mathbf{\Pi}\geq 0} \langle \mathbf{C}, \mathbf{\Pi}\rangle + \varepsilon\,\mathrm{KL}\left(\mathbf{\Pi}\,\|\,\mathbf{a}\mathbf{b}^\top\right) \tag{18}$$

where the resulting transport plan $\mathbf{\Pi}$ is obtained using Sinkhorn iterations.

The entropic regularizer yields smooth and differentiable assignments and improves numerical stability, while the approximately doubly stochastic constraints promote balanced expert utilization and mitigate expert collapse. After estimating $\mathbf{\Pi}$, tokens are softly routed to experts, each expert head produces class logits, and the expert outputs are aggregated back to token locations using $\mathbf{\Pi}$ to obtain token-wise logits. Finally, token-wise logits are reshaped to the corresponding spatial lattice and upsampled when needed to form the per-pixel classification map, followed by a softmax to obtain per-pixel class probabilities.

IV. Experimental Results and Analysis

A. Data Description

To demonstrate the effectiveness of DRIDNet, we evaluate it on four widely used multisensor remote sensing image

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TABLE I
DATASET DESCRIPTION

Attribute	Houston2013	Houston2018	MUUFL Gulfport	Trento
Location	Houston, Texas, USA	Greater Houston, Texas, USA	Gulfport, Mississippi, USA	South of Trento, Italy
Modalities	HSI, LiDAR	HSI, LiDAR	HSI, LiDAR	HSI, LiDAR
HSI Image Size	349×1905	1202×4172	325×220	166×600
HSI Spatial Resolution	2.5 m	1.0 m	1.0 m	1.0 m
LiDAR Spatial Resolution	2.5 m	0.5 m	1.0 m	1.0 m
HSI Bands	144	48	64	63
HSI Wavelength Range	0.38 to 1.05 µm	0.38 to 1.05 µm	0.375 to 1.05 µm	0.403 to 0.989 µm
HSI Sensor	ITRES CASI-1500	ITRES CASI-1500	ITRES CASI-1500	AISA Eagle
LiDAR Sensor	NCALM airborne LiDAR	Optech Titan MW	Gemini LiDAR	Optech ALTM 3100EA

TABLE II
INFLUENCE OF PCA PLACEMENT ON HOUSTON2013

Strategy	OA (%)	AA (%)	Kappa (%)	Params (M)	GFLOPs (per sample)	Test time (s)
No PCA	91.14	91.95	90.38	10.94	0.3014	5.170
PCA before CIIDM	90.69	91.91	89.90	2.44	0.0321	3.151
PCA after CIIDM	94.31	94.92	93.83	2.44	0.0326	4.826

classification datasets, including Houston2013, Houston2018, MUUFL Gulfport, and Trento. Table I summarizes the key characteristics of these datasets. To avoid ambiguity in the evaluation protocol, we clarify that different datasets follow their commonly adopted benchmark or limited-label protocols. For Houston2013, we follow the official train/test split. For Houston2018, we follow the benchmark protocol widely adopted in recent HSI–LiDAR and multisource remote sensing classification studies. For MUUFL, where no unique official train/test split is universally enforced, we follow the commonly used limited-label protocol with 100 training samples per class. For Trento, considering that the conventional fixed split has become nearly saturated in recent studies, we adopt a more challenging limited-label protocol with 20 training samples per class. All compared methods are evaluated under exactly the same data partitions for fair comparison.

1) Houston2013: The Houston2013 dataset was acquired in 2012 over the University of Houston campus and adjacent urban areas. It provides co-registered hyperspectral imagery with 144 bands covering 380–1050 nm and an LiDAR-derived digital surface model at 2.5-m ground sampling distance. The image size is 349 × 1905 pixels. The dataset contains 15 land cover classes and 15 029 labeled pixels.

2) Houston2018: The Houston2018 dataset is released by the 2018 IEEE GRSS Data Fusion Contest for multimodal urban classification. It includes LiDAR point clouds at 1550, 1064, and 532 nm, from which intensity rasters and digital surface models at 0.5-m ground sampling distance are derived. It also provides hyperspectral imagery with 48 bands covering 380–1050 nm at 1-m ground sampling distance and RGB imagery at 5-cm ground sampling distance. The ground truth contains 20 classes.

3) MUUFL Gulfport: The MUUFL Gulfport dataset was acquired in November 2010 over the University of Southern Mississippi Gulf Park campus in Gulfport, Mississippi. It provides co-registered hyperspectral imagery and LiDAR products including intensity and a digital surface model. We use a 325 × 220 region with 64 spectral bands and LiDAR elevation and intensity channels at 1-m ground sampling distance. The ground truth defines 11 classes.

4) Trento: The Trento dataset was collected over a rural area south of Trento, Italy. It provides hyperspectral imagery with 63 bands covering approximately 0.40–0.99 μm and a co-registered LiDAR-derived digital surface model at about 1-m ground sampling distance. The image size is 166×600 pixels. The ground truth includes six classes with 30 214 labeled pixels.

B. Implementation Details

The proposed framework is implemented in PyTorch and trained on a workstation equipped with NVIDIA GeForce RTX 3090 GPUs. Unless otherwise specified, all experiments are trained for 100 epochs with a mini-batch size of 64. The network parameters are optimized using Adam with an initial learning rate of 2×10^{-4}. During training, the learning rate is reduced adaptively according to the overall accuracy (OA), with a decay factor of 0.5 and a patience of 5 epochs. For settings involving random sample selection, we repeat the experiments using multiple random seeds and report the mean ± standard deviation of OA, average accuracy (AA), and the Kappa coefficient. For each dataset, all the methods compared are evaluated using exactly the same partitions under each seed. The classification branch is supervised by the cross-entropy loss, and no explicit weight decay is used in the reported experiments.

The intrinsic decomposition branch is optimized jointly with the downstream classification framework. In the decomposition module, Fourier feature encoding with four frequency bands is used to represent spatial–spectral coordinates, and the coordinate implicit predictor is implemented by a multilayer perceptron with hidden width 128 and four hidden layers. For the spectral prior, the 1-D U-Net adopts a base width of 128, and the diffusion process follows a cosine noise schedule with 1000 timesteps. These SDDP-related hyperparameters are selected to provide sufficient capacity for modeling interband reflectance correlations while keeping the training-time regularization cost affordable. During joint training, a subset of at most 2048 pixel-wise reflectance vectors is sampled from each mini-batch to compute the prior regularization term, which

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TABLE III
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

No.	Class (Train/Test)	FusAtNet	CALC	HI2D2FNet	MIViT	FDNet	M2FNet	HyperDID	S3F2Net	DCMNet	MCFNet	E-Mamba	DRIDNet
1	Healthy grass (198/1053)	83.20	81.89	85.83	82.75	83.18	**94.76**	82.88	83.13	83.27	82.91	86.91	83.03
2	Stressed grass (190/1064)	81.68	79.54	83.46	85.09	83.82	84.85	99.69	85.18	84.67	82.05	**99.83**	99.19
3	Synthetic grass (192/505)	**100.00**	96.77	99.61	99.10	99.30	97.32	96.13	95.60	99.21	95.01	99.60	97.37
4	Trees (188/1056)	92.52	94.72	99.23	93.22	95.25	97.14	91.45	97.85	92.12	93.19	95.09	**99.36**
5	Soil (186/1056)	98.87	99.93	92.13	99.94	99.89	99.32	99.98	98.14	99.98	99.06	99.92	**100.00**
6	Water (182/143)	95.81	95.74	95.03	95.66	95.90	95.90	94.51	96.02	**99.25**	98.60	95.85	99.18
7	Residential (196/1072)	88.81	89.86	90.95	91.17	88.14	89.54	92.05	92.57	**96.09**	88.81	90.96	92.58
8	Commercial (191/1053)	81.96	79.04	82.03	93.67	77.76	88.02	87.63	**95.85**	82.79	82.25	91.37	94.71
9	Road (193/1059)	83.48	90.96	77.33	**98.81**	83.18	92.53	83.92	92.10	88.27	84.33	83.58	93.13
10	Highway (191/1036)	45.75	66.24	39.93	67.89	61.66	69.96	90.42	84.11	74.86	63.04	**92.00**	89.39
11	Railway (181/1054)	81.12	96.89	61.28	**99.85**	86.70	89.83	85.46	97.09	97.32	82.64	94.04	89.68
12	Parking lot 1 (192/1041)	95.78	96.66	66.63	92.54	89.99	93.74	87.58	98.02	98.32	91.17	**98.58**	97.52
13	Parking lot 2 (184/285)	**93.01**	86.26	56.92	90.05	89.92	89.92	91.68	89.66	91.67	91.91	85.29	89.86
14	Tennis court (181/247)	**100.00**	99.94	99.34	99.87	94.43	**100.00**	96.05	93.72	99.35	99.59	**100.00**	99.29
15	Running track (187/473)	99.83	97.83	96.40	99.68	**100.00**	**100.00**	99.24	**100.00**	99.58	99.54	99.37	99.52
OA (%)(2832/12197)		85.41 ±0.42	88.69 ±0.36	79.74 ±0.68	91.48 ±0.31	86.61 ±0.44	90.96 ±0.33	90.93 ±0.37	92.84 ±0.28	90.91 ±0.34	86.58 ±0.46	93.71 ±0.25	**94.31 ±0.22**
AA (%)		88.12 ±0.39	90.15 ±0.34	81.74 ±0.71	92.62 ±0.30	88.61 ±0.41	92.19 ±0.32	91.91 ±0.35	93.27 ±0.29	92.45 ±0.31	88.94 ±0.43	94.16 ±0.27	**94.92 ±0.21**
Kappa (×100)		84.19 ±0.46	87.75 ±0.40	78.02 ±0.74	90.80 ±0.34	85.51 ±0.48	90.21 ±0.36	90.15 ±0.40	92.24 ±0.30	90.15 ±0.37	85.49 ±0.50	93.18 ±0.27	**93.83 ±0.24**

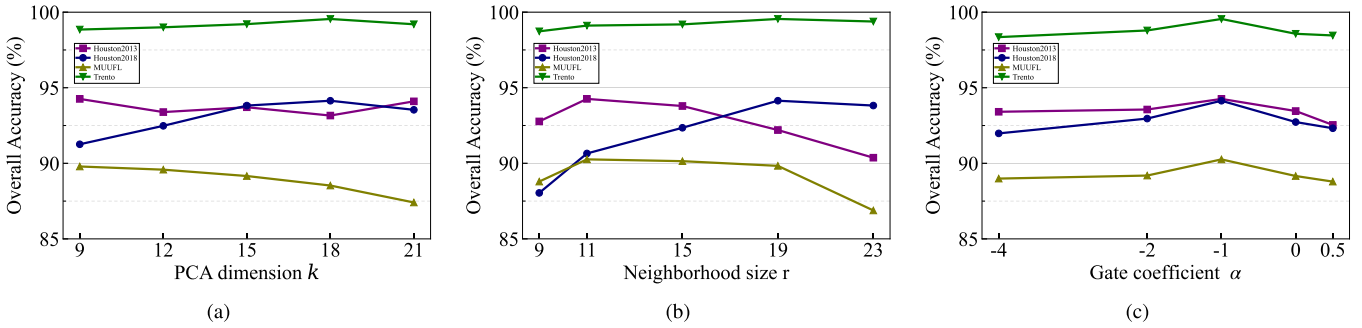


Fig. 5. Parameter sensitivity analysis of the proposed method in terms of validation OA. (a) Effect of the PCA dimension k. (b) Effect of the neighborhood size r. (c) Effect of the gate coefficient α.

provides a stable compromise between training efficiency and spectral regularization strength.

In the CTRM, the latent prototype dimension is set to 64, and the transport assignment is computed with seven Sinkhorn iterations. These routing-related hyperparameters are selected to balance routing capacity, numerical stability, and computational efficiency. The routing temperature is treated as a tunable hyperparameter and selected on the validation split. For reflectance reinjection, the residual mixing coefficient is defined as $g = \sigma(\alpha)$, where the initial value of α is determined according to the parameter sensitivity analysis.

C. Parameter Sensitivity Analysis

All hyperparameters are selected using a held-out validation subset derived only from the training split of each dataset, while the test split is used only once for the final performance report after all hyperparameters are fixed. Accordingly, the curves in Fig. 5 report validation OA. We study the sensitivity of DRIDNet to three hyperparameters: the PCA dimension k, the neighborhood size r, and the gate coefficient α, while keeping all other settings fixed. These parameters control the retained dimensionality of the PCA-projected hyperspectral representation after reflectance-gated residual reinjection, the spatial context used in patch-based processing, and the reflectance-residual reinjection strength in CIIDM, respectively.

1) Influence of PCA Placement: Before analyzing the retained PCA dimension, we first compare three PCA placement strategies on Houston2013, as reported in Table II. Although removing PCA keeps all original spectral bands, it leads to much higher model complexity. Applying PCA before CIIDM reduces the computational cost but yields the lowest OA, suggesting that early spectral compression may remove information useful for intrinsic decomposition. In contrast, applying PCA after CIIDM achieves the best OA/AA/Kappa while maintaining a compact model. Therefore, PCA is applied after reflectance-gated residual reinjection in the final setting.

2) Influence of PCA Dimension k: The PCA dimension k determines the dimensionality of the postreinjection hyperspectral representation used by the downstream encoder and therefore trades spectral discriminability against redundancy and noise. As shown in Fig. 5(a), the effect of k is dataset dependent. On Houston2013, the validation OA changes marginally over the tested range, indicating low sensitivity to k. On Houston2018 and Trento, the validation OA improves when increasing k from small to moderate values and then

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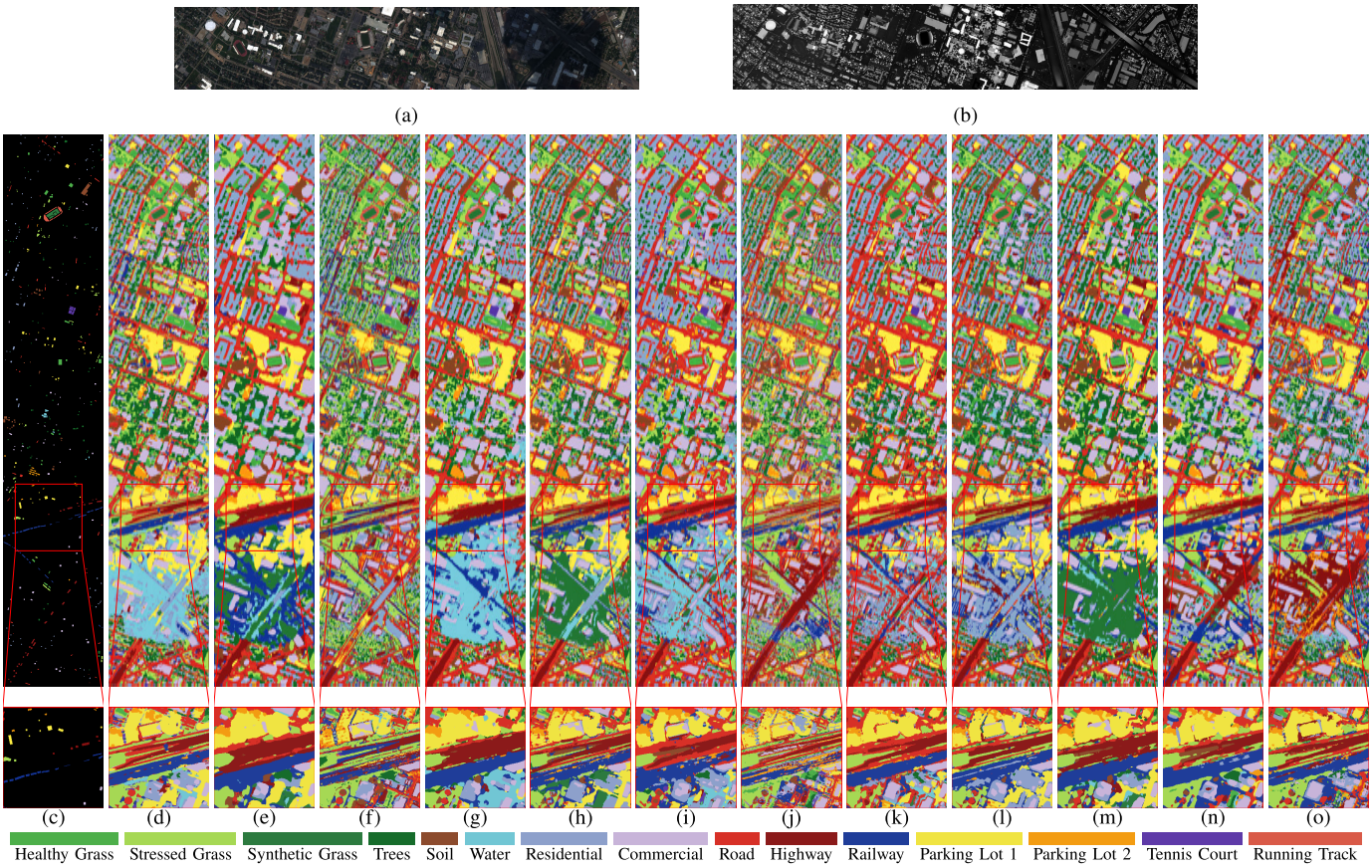


Fig. 6. Classification maps of different methods for the Houston2013 dataset. (a) Pseudocolor image for HSI. (b) LiDAR-based DSM. (c) Ground truth map. (d) FusAtNet (85.41%). (e) CALC (88.69%). (f) HI2D2FNet (79.74%). (g) MIViT (91.48%). (h) FDNet (86.61%). (i) M2FNet (90.96%). (j) HyperDID (90.93%). (k) S3F2Net (92.84%). (l) DCMNet (90.91%). (m) MCFNet (86.58%). (n) E-Mamba (93.71%). (o) DRIDNet (94.31%).

saturates or slightly degrades, suggesting that a moderate spectral subspace is sufficient. In contrast, MUUFL achieves its best validation OA at a relatively small k and deteriorates as k increases, implying that excessive spectral dimensionality may introduce redundant or noisy components under limited spatial extent and supervision. Based on these validation trends, we set $k = 21$ for Houston2013, $k = 18$ for Houston2018, $k = 9$ for MUUFL, and $k = 18$ for Trento in the final experiments.

3) Influence of Neighborhood Size r: The neighborhood size r defines the spatial support of paired patches, affecting both the local coordinate range for CIIDM and the effective spatial context for subsequent feature extraction. Fig. 5(b) shows that Houston2013 and MUUFL peak at small to moderate neighborhoods and then decline as r increases, indicating that overly large neighborhoods can dilute fine structures and blur class boundaries. By contrast, Houston2018 benefits consistently from larger neighborhoods within the tested range, while Trento reaches its maximum at a moderately large neighborhood and remains comparable at the largest setting. Overall, scenes with broader homogeneous regions favor larger spatial context, whereas scenes with sharper boundaries prefer smaller neighborhoods. Based on the validation results, we set $r = 11$ for Houston2013, $r = 19$ for Houston2018, $r = 11$ for MUUFL, and $r = 19$ for Trento in the final experiments.

4) Influence of Gate Coefficient α: The gate coefficient α controls the reflectance residual reinjection in (5) through the gating function $\sigma(\alpha)$, and thus balances the contribution of the recovered reflectance R and the original radiance observation. As shown in Fig. 5(c), a consistent trend is observed across all datasets. The validation OA increases from strongly negative α to a moderately negative value and reaches the maximum at $\alpha = -1$, then decreases as α approaches zero or becomes positive. This indicates that a moderately negative gate yields the most effective tradeoff between leveraging intrinsic decomposition and avoiding error propagation from imperfect decomposition. Therefore, we adopt $\alpha = -1$ for all four datasets in the final experiments.

D. Comparison and Analysis of Classification Performance

To validate the effectiveness of DRIDNet, we compare it with representative state-of-the-art methods using both quantitative evaluation and visual inspection of classification maps. Following standard practice, we report OA, AA, and the kappa coefficient (Kappa). These three metrics provide complementary assessments of per-pixel predictions against the ground truth labels.

The baselines include FusAtNet [50], CALC [51], HI2D2FNet [24], MIViT [52], FDNet [29], M2FNet [53], HyperDID [45], S3F2Net [28], DCMNet [54], MCFNet [55], and E-Mamba [36]. Tables III–VI summarize per-class accuracies together with OA, AA, and Kappa on the four datasets. The corresponding classification maps are shown in Figs. 6–9.

Overall, CNN-based and frequency domain fusion methods such as FusAtNet, FDNet, and M2FNet deliver strong

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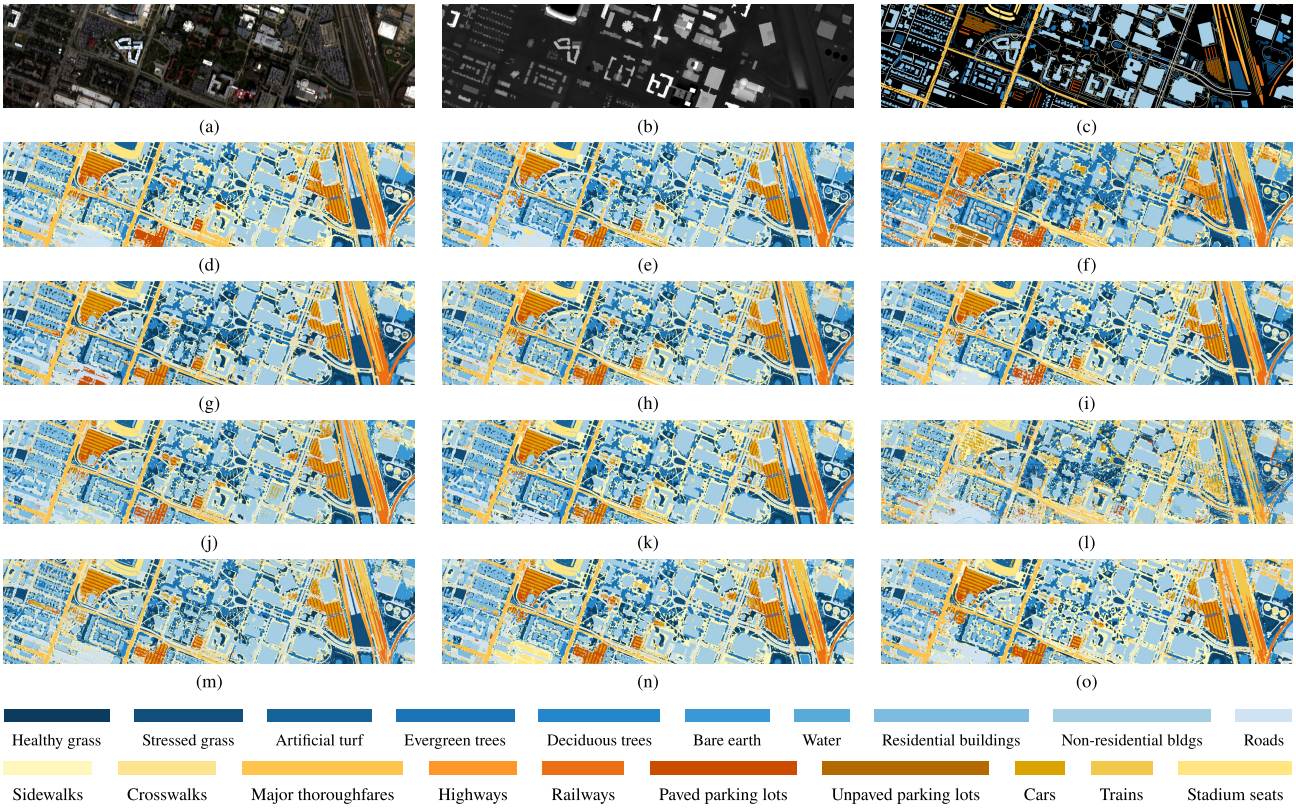


Fig. 7. Classification maps of different methods for the Houston2018 dataset. (a) Pseudocolor image for HSI. (b) LiDAR-based DSM. (c) Ground truth map. (d) FusAtNet (91.01%). (e) CALC (91.42%). (f) HI2D2FNet (72.76%). (g) MIViT (91.26%). (h) FDNet (92.48%). (i) M2FNet (90.23%). (j) HyperDID (90.42%). (k) S3F2Net (91.48%). (l) DCMNet (92.91%). (m) MCFNet (89.55%). (n) E-Mamba (93.62%). (o) DRIDNet (94.06%).

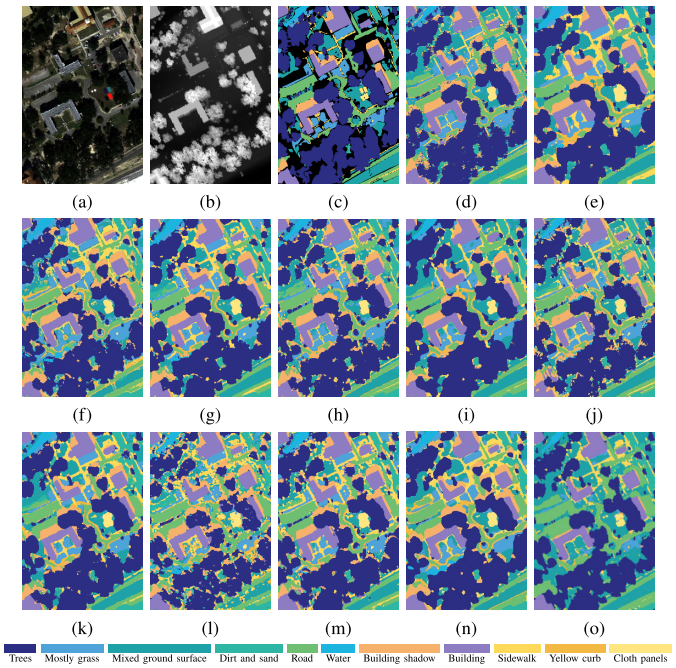


Fig. 8. Classification maps of different methods for the MUUFL dataset. (a) Pseudocolor image of HSI. (b) LiDAR-derived DSM. (c) Ground-truth map. (d) FusAtNet (OA = 88.89%). (e) CALC (OA = 87.73%). (f) HI2D2FNet (OA = 84.21%). (g) MIViT (OA = 88.26%). (h) FDNet (OA = 87.96%). (i) M2FNet (OA = 89.12%). (j) HyperDID (OA = 88.72%). (k) S3F2Net (OA = 87.63%). (l) DCMNet (OA = 88.83%). (m) MCFNet (OA = 87.71%). (n) E-Mamba (OA = 88.13%). (o) DRIDNet (OA = 90.43%).

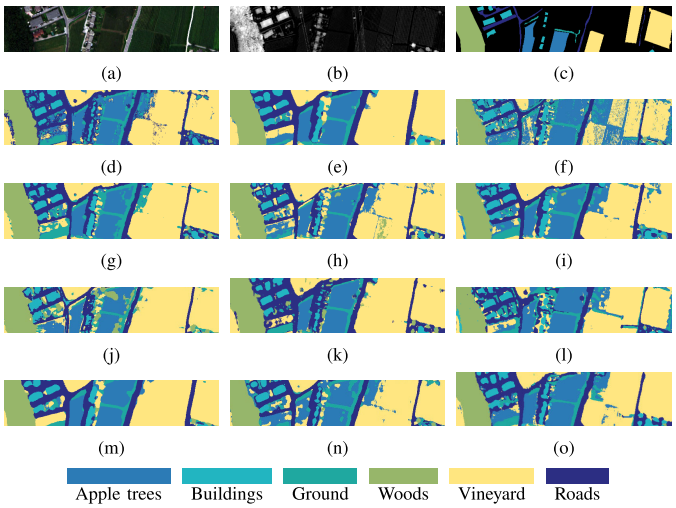


Fig. 9. Classification maps of different methods for the Trento dataset. (a) Pseudocolor image of HSI. (b) LiDAR-derived DSM. (c) Ground-truth map. (d) FusAtNet (OA = 98.28%). (e) CALC (OA = 98.04%). (f) HI2D2FNet (OA = 91.88%). (g) MIViT (OA = 98.48%). (h) FDNet (OA = 98.94%). (i) M2FNet (OA = 98.27%). (j) HyperDID (OA = 97.20%). (k) S3F2Net (OA = 98.26%). (l) DCMNet (OA = 96.91%). (m) MCFNet (OA = 98.33%). (n) E-Mamba (OA = 97.53%). (o) DRIDNet (OA = 99.32%).

performance on classes with stable appearance and clear spatial structure. CALC and MIViT further improve the modeling of complex urban patterns and long-range dependencies in the Houston scenes. S3F2Net, DCMNet, and MCFNet strengthen cross-modal interactions and generally achieve the best results among existing baselines. Despite these advances, the competing approaches operate directly on observed radiance and adopt fixed fusion designs without explicit control of intrinsic

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TABLE IV
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE HOUSTON2018 DATASET

No.	Class (Train/Test)	FusAtNet	CALC	HI2D2FNet	MIViT	FDNet	M2FNet	HyperDID	S3F2Net	DCMNet	MCFNet	E-Mamba	DRIDNet
1	Healthy grass (1000/38196)	97.46	96.37	94.07	**99.84**	96.73	95.49	94.22	94.71	94.21	97.72	98.14	92.36
2	Stressed grass (1000/129008)	90.54	90.77	82.17	70.12	**93.58**	91.86	92.31	93.09	93.21	91.49	91.91	92.63
3	Artificial turf (1000/1736)	99.96	99.98	99.84	**99.99**	99.97	99.98	99.93	99.98	99.97	99.91	99.95	99.98
4	Evergreen trees (1000/53322)	97.52	98.93	94.31	99.35	99.02	98.77	99.25	**99.49**	97.81	98.86	98.79	98.81
5	Deciduous trees (1000/19172)	98.10	**99.43**	87.71	93.08	98.92	97.49	98.76	99.39	99.24	87.33	98.64	98.90
6	Bare earth (1000/17064)	99.62	99.97	98.58	99.96	99.89	99.69	99.95	99.75	99.98	99.86	99.97	**99.99**
7	Water (500/564)	99.94	99.95	98.82	99.96	99.71	99.93	99.86	99.97	99.95	**99.98**	99.91	99.96
8	Residential buildings (1000/157995)	94.12	95.38	87.88	97.41	96.23	92.10	96.52	97.03	96.59	88.16	96.61	**98.18**
9	Non-residential buildings (1000/893769)	94.18	95.22	82.45	96.18	94.77	95.39	91.12	95.11	95.07	92.97	**96.24**	95.28
10	Roads (1000/182283)	73.08	**84.96**	26.34	81.47	79.03	69.68	76.73	63.04	82.16	82.52	83.01	82.61
11	Sidewalks (1000/135035)	80.86	71.89	46.32	71.62	81.28	80.35	79.02	85.19	82.88	76.62	82.71	**85.31**
12	Crosswalks (1000/5059)	86.31	85.09	67.51	98.69	95.82	91.08	95.74	97.12	98.01	97.08	**98.81**	96.48
13	Major thoroughfares (1000/184438)	88.92	79.04	38.06	88.97	88.11	76.83	88.24	89.18	87.72	75.48	88.59	**95.74**
14	Highways (1000/38438)	84.47	97.93	84.02	99.05	98.81	98.68	99.29	98.74	98.80	98.04	**99.42**	98.86
15	Railways (1000/26748)	98.72	98.04	97.70	99.78	99.75	99.47	**99.91**	99.72	99.83	94.63	99.70	99.69
16	Paved parking lots (1000/44932)	98.55	97.51	89.51	93.18	98.22	95.43	98.66	94.98	98.02	97.64	98.88	**99.58**
17	Unpaved parking lots (250/337)	99.88	99.93	99.47	99.96	99.91	99.95	99.83	99.92	99.94	99.58	99.89	**99.97**
18	Cars (1000/25289)	98.02	98.76	88.73	**99.34**	98.62	97.72	98.70	99.11	98.94	98.61	98.82	99.04
19	Trains (1000/20479)	99.82	99.51	97.89	99.96	99.84	99.69	99.91	99.93	99.78	99.37	99.95	**99.97**
20	Stadium seats (1000/26296)	99.74	99.83	98.15	99.88	99.86	99.82	99.91	99.91	**99.96**	99.86	99.76	99.95
OA (%)(18750/2000160)		91.01 ±0.31	91.42 ±0.28	72.76 ±0.72	91.26 ±0.26	92.48 ±0.23	90.23 ±0.34	90.42 ±0.34	91.48 ±0.27	92.91 ±0.24	89.55 ±0.36	93.62 ±0.25	**94.06 ±0.18**
AA (%)		93.99 ±0.25	94.42 ±0.21	82.98 ±0.60	94.39 ±0.24	95.90 ±0.19	93.97 ±0.29	95.39 ±0.27	95.27 ±0.23	96.10 ±0.20	93.79 ±0.30	96.49 ±0.22	**96.66 ±0.15**
Kappa (×100)		88.42 ±0.36	88.92 ±0.33	66.05 ±0.84	88.67 ±0.31	90.28 ±0.28	87.37 ±0.40	87.72 ±0.43	89.02 ±0.32	90.84 ±0.30	86.58 ±0.42	91.74 ±0.32	**92.32 ±0.23**

decomposition or routing. This limits robustness under illumination variations and shadows and reduces stability in small sample regimes. In contrast, DRIDNet achieves the best or near-best OA, AA, and Kappa across all four datasets.

1) Performance Comparison on Houston2013: Houston2013 includes diverse natural and urban categories and is affected by notable illumination disturbances such as clouds and shadows. As reported in Table III, DRIDNet attains the highest or near-highest accuracies for most categories and remains among the top performers for almost all classes. The improvements are pronounced for land cover types with similar spectra and ambiguous boundaries, as well as for urban regions with complex spatial layouts.

The classification maps in Fig. 6 show that DRIDNet reduces errors in shadow-affected areas, suppresses isolated noisy predictions within homogeneous regions, and yields more continuous boundaries. These trends are consistent with the use of intrinsic decomposition and spectral regularization for reducing illumination-induced variability, together with multiscale feature encoding and cross-modal routing for preserving material cues and structural information after fusion.

2) Performance Comparison on Houston2018: Houston2018 is a large-scale urban benchmark with many categories and intricate man-made structures. Table IV indicates that several methods already achieve high accuracies on classes with clear shapes and regular outlines, while larger performance gaps remain on categories with strong spatial interleaving and complex interfaces. DRIDNet yields higher accuracy on these challenging regions and reduces confusion between adjacent classes.

Fig. 7 further shows that DRIDNet produces more spatially coherent predictions in complex urban regions. It better preserves elongated road structures, yields cleaner boundaries between adjacent classes, and suppresses scattered noisy predictions in relatively homogeneous areas. These qualitative results are consistent with the quantitative comparisons and indicate that the proposed framework improves structural consistency and class discrimination in large-scale urban scenes.

3) Performance Comparison on MUUFL Gulfport: MUUFL has limited labeled samples and strong spectral similarity among several classes. It also exhibits substantial intraclass variability and a long-tail label distribution. Table V shows that many methods approach saturation on easier categories, whereas differences remain visible on classes with scarce labels and complex backgrounds. DRIDNet achieves higher OA and AA than competing approaches and maintains stronger recognition for multiple minority classes.

Fig. 8 further confirms that the proposed framework suppresses isolated errors in transition regions and produces more contiguous class areas near fine boundaries and small targets. This behavior is consistent with the stabilizing effect of physically grounded decomposition and the robustness brought by multiscale cross-modal alignment and routing under limited supervision.

4) Performance Comparison on Trento: Trento consists of rural and urban transition scenes with large continuous land cover regions. The number of training samples per class is very small, which makes generalization challenging. As reported in Table VI, most methods already perform strongly on this dataset, yet DRIDNet still achieves the best OA, AA, and Kappa and remains competitive across all classes.

The visual comparisons in Fig. 9 show that the proposed method produces clean and spatially consistent predictions in orchards and forests and preserves elongated structures such

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TABLE V
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE MUUFL DATASET

No.	Class (Train/Test)	FusAtNet	CALC	HI2D2FNet	MIViT	FDNet	M2FNet	HyperDID	S3F2Net	DCMNet	MCFNet	E-Mamba	DRIDNet
1	Trees (100/23146)	92.87	91.12	91.86	89.28	90.55	92.29	91.37	86.94	91.61	89.72	91.95	**93.12**
2	Mostly grass (100/4170)	80.42	77.76	83.02	**90.21**	88.66	78.52	82.47	81.95	87.42	85.74	83.28	89.32
3	Mixed ground surface (100/6782)	75.48	81.32	56.97	77.36	72.83	82.46	78.84	**82.91**	78.24	69.93	76.25	77.54
4	Dirt and sand (100/1726)	94.12	89.62	84.08	93.47	92.10	89.68	93.68	**94.28**	93.05	90.94	92.36	90.53
5	Road (100/6587)	87.16	77.91	79.84	86.59	86.61	**90.05**	88.12	88.20	84.58	87.47	83.92	88.55
6	Water (100/366)	81.64	99.82	99.18	99.91	98.75	99.54	96.31	99.87	99.56	**99.94**	99.82	99.18
7	Building shadow (100/2133)	**96.78**	93.91	91.26	96.35	95.42	90.50	95.04	96.73	92.84	96.51	93.48	96.62
8	Building (100/6140)	95.14	**97.34**	93.74	91.31	91.86	94.82	91.10	93.04	94.25	96.08	94.22	94.41
9	Sidewalk (100/1285)	75.81	79.28	54.62	**89.86**	84.72	61.95	84.02	84.74	77.89	88.76	70.81	89.46
10	Yellow curb (100/83)	85.33	93.61	76.45	94.38	95.43	98.36	97.24	98.08	97.28	95.54	96.02	**99.82**
11	Cloth panels (100/169)	96.03	98.57	91.28	97.08	95.49	96.64	98.55	**99.22**	98.55	98.49	95.57	98.58
OA (%)(1100/52587)		88.89 ±0.36	87.73 ±0.39	84.21 ±0.68	88.26 ±0.34	87.96 ±0.41	89.12 ±0.32	88.72 ±0.44	87.63 ±0.37	88.83 ±0.35	87.71 ±0.42	88.13 ±0.48	**90.43 ±0.22**
AA (%)		87.34 ±0.33	89.11 ±0.31	82.03 ±0.62	91.44 ±0.29	90.22 ±0.35	88.62 ±0.30	90.61 ±0.42	91.45 ±0.34	90.48 ±0.28	90.83 ±0.37	88.88 ±0.45	**92.44 ±0.19**
Kappa (×100)		86.67 ±0.41	84.05 ±0.44	79.42 ±0.75	85.55 ±0.38	84.38 ±0.47	**87.92 ±0.37**	85.08 ±0.50	87.31 ±0.39	85.49 ±0.40	83.97 ±0.45	84.11 ±0.54	87.48 ±0.25

TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE TRENTO DATASET

No.	Class (Train/Test)	FusAtNet	CALC	HI2D2FNet	MIViT	FDNet	M2FNet	HyperDID	S3F2Net	DCMNet	MCFNet	E-Mamba	DRIDNet
1	Apple trees (20/4014)	98.72	98.78	84.72	97.68	**99.82**	97.36	99.32	99.01	98.33	98.29	87.92	98.58
2	Building (20/2883)	93.82	97.12	96.84	98.62	97.42	97.58	85.74	96.58	96.31	97.71	95.48	**99.15**
3	Ground (20/459)	97.55	95.70	92.48	98.45	96.41	**99.34**	96.26	97.02	93.74	97.26	97.62	96.48
4	Woods (20/9103)	98.28	98.36	99.86	98.48	99.68	99.25	99.87	98.61	99.16	98.62	**99.92**	99.91
5	Vineyard (20/10481)	99.36	98.57	85.47	99.08	99.81	98.61	**99.84**	98.96	94.82	98.97	99.74	99.65
6	Roads (20/3154)	**98.36**	95.61	94.61	97.36	94.52	95.94	88.62	95.73	96.52	96.14	97.36	98.04
OA (%)(120/30094)		98.28 ±0.18	98.04 ±0.22	91.88 ±0.72	98.48 ±0.16	98.94 ±0.13	98.27 ±0.20	97.20 ±0.46	98.26 ±0.19	96.91 ±0.31	98.33 ±0.18	97.53 ±0.42	**99.32 ±0.09**
AA (%)		97.68 ±0.21	97.36 ±0.25	92.33 ±0.68	98.28 ±0.18	97.94 ±0.20	98.01 ±0.22	94.94 ±0.51	97.65 ±0.24	96.48 ±0.35	97.83 ±0.23	96.34 ±0.48	**98.64 ±0.12**
Kappa (×100)		97.71 ±0.24	97.82 ±0.27	89.02 ±0.81	98.22 ±0.21	98.57 ±0.17	97.68 ±0.25	96.35 ±0.55	98.12 ±0.23	95.89 ±0.38	98.06 ±0.25	96.72 ±0.50	**99.08 ±0.11**

as roads. The results indicate that the proposed representation remains stable under very limited annotations and that the fusion design captures multiscale geometry and cross-modal relations effectively.

E. Shadow Region Evaluation on Houston2013

In order to evaluate robustness under severe illumination variation, the shadow mask is generated only from the band-wise normalized HSI, without using test labels or model predictions. A shadow detector is applied to obtain a pixel-wise shadow probability map $P_s(x, y) \in [0, 1]$, and the binary shadow mask is defined as $M_s(x, y) = \mathbf{1}(P_s(x, y) > \tau_s)$, where τ_s is fixed before comparing all methods. Morphological cleanup and connected-component filtering are then used to remove isolated responses and obtain the final shadow regions, from which a representative hard local shadow patch is selected.

Based on the final mask M_s, labeled pixels inside the mask are used for the full shadow-region evaluation, while labeled pixels outside the mask are used for the full nonshadow-region evaluation. This partition is used only for evaluation and does not change the training samples.

The qualitative comparison in Fig. 10 is consistent with the quantitative analysis. Compared with competing methods, the proposed method produces fewer error pixels and clearer spatial continuity in the shadow patch, particularly around thin linear structures and small scattered objects. This observation agrees with the consistent gains observed on both the full shadow region and the local shadow patch.

The improvement can be attributed to the synergy of the proposed modules. CIIDM reduces the distribution shift by disentangling reflectance-related cues from illumination variation. SDDP regularizes spectral estimation in under-exposed regions and mitigates noise amplification as well as spectral distortion. DPFEM and CTRM enable adaptive fusion between hyperspectral imagery and LiDAR, so that structural cues can compensate when spectral evidence becomes unreliable in shadows. Together, these design choices yield consistently better performance in shadowed regions while maintaining competitive accuracy in nonshadow areas.

Fig. 11 compares the OA on the full shadow and full nonshadow regions. On the full shadow region, DRIDNet yields the best performance and improves the strongest baseline by 0.65 percentage points. The margin over representative competitors is substantially larger, reaching 7.40 and 8.88 percentage points against MIViT and M2FNet, respectively. This advantage is not limited to OA. In the shadow region, the proposed method increases AA by 4.96 percentage points over the strongest baseline and improves Kappa by 0.018. It also provides a higher mIoU than the best competing method, indicating more reliable spatial agreement under shadow-induced radiometric distortion.

On the full nonshadow region, DRIDNet remains competitive and slightly improves the strongest baseline by

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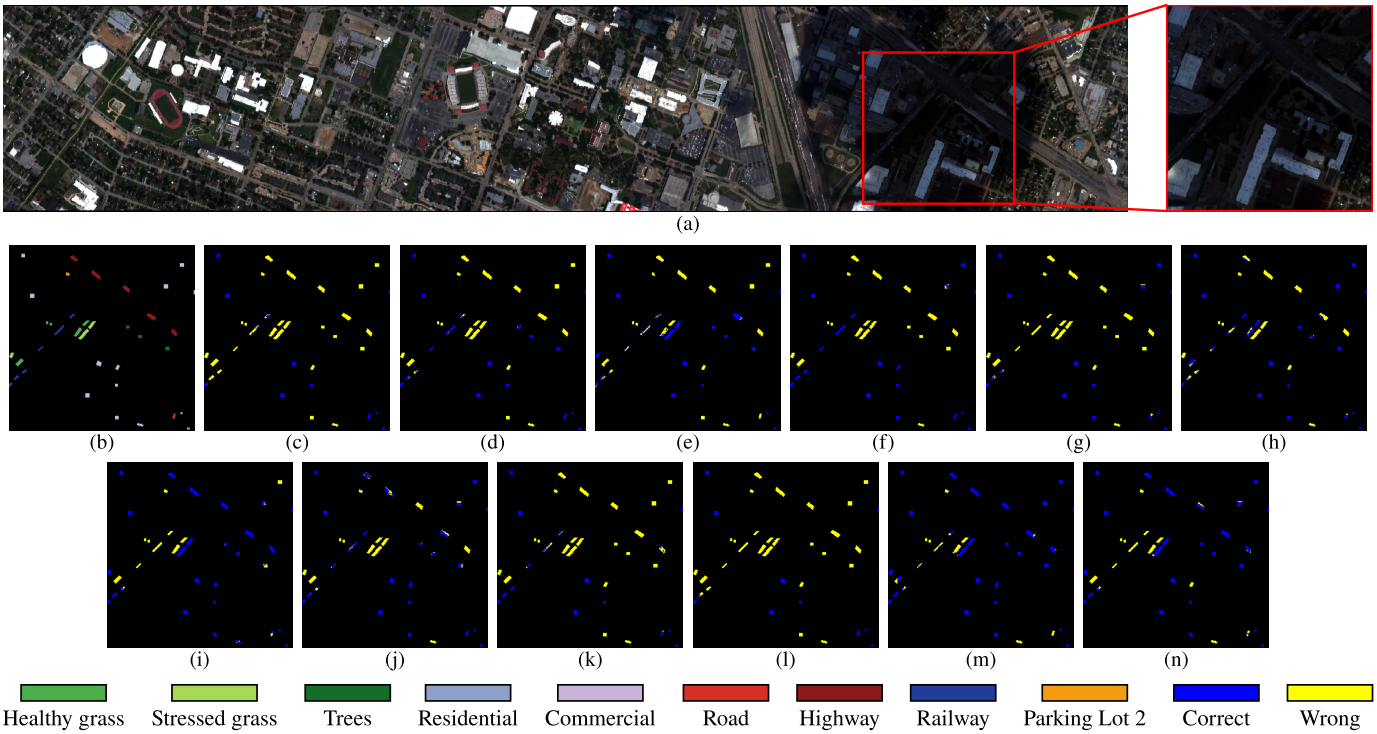


Fig. 10. Qualitative comparison on a representative shadow patch of Houston2013. (a) Pseudo RGB visualization of the selected patch. (b) Ground truth. (c) FusAtNet. (d) CALC. (e) HI2D2FNet. (f) MIViT. (g) FDNet. (h) M2FNet. (i) HyperDID. (j) S3F2Net. (k) DCMNet. (l) MCFNet. (m) E-Mamba. (n) DRIDNet. For (c)–(n), blue indicates correct predictions and yellow indicates errors with respect to the ground truth.

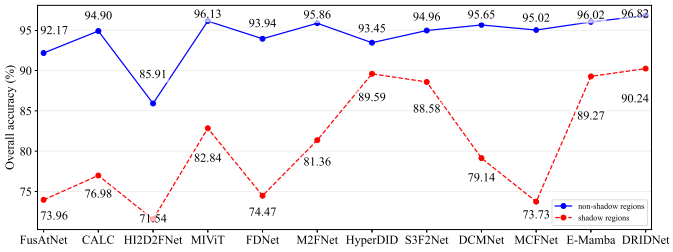


Fig. 11. OA comparison on the Houston2013 full shadow and full nonshadow labeled regions.

0.69 percentage points in OA. More importantly, it substantially reduces shadow sensitivity. The OA drop from nonshadow to shadow is 6.58 percentage points for the proposed method, which is markedly smaller than the drops of major competitors. Compared with MIViT, FDNet, and MCFNet, DRIDNet reduces the shadow-induced OA degradation by 6.71, 12.89, and 14.71 percentage points, respectively. These results indicate that DRIDNet not only preserves strong performance in well-lit areas, but also narrows the accuracy gap between shadowed and nonshadowed conditions.

The local patch evaluation further examines fine-grained performance in hard shadows and confirms the robustness gain. On the selected representative shadow patch, the proposed method improves the strongest baseline by 14.71 percentage points in OA.

F. T-SNE Visualization and Feature Discriminability

Finally, Fig. 12 visualizes the feature embeddings from the penultimate layer on Houston2013 using t-SNE. For CNN-based and transformer-based fusion baselines, the embeddings exhibit noticeable interclass overlap, especially among spectrally similar vegetation categories and between built-up regions and road-like classes; moreover, several classes form fragmented and scattered clusters. Stronger interaction-based models such as S3F2Net, DCMNet, and MCFNet produce more compact manifolds, but class-boundary ambiguity remains, and minority categories are partly merged into dominant clusters. By contrast, DRIDNet produces more separable and internally coherent clusters for most categories, with clearer margins for rare classes. These observations suggest that the intrinsic decomposition and diffusion-based spectral regularization, together with cross-modal routing, promote a more discriminative and semantically aligned representation space, which is consistent with the quantitative gains reported in the preceding experiments.

G. Computational Complexity Analysis

Model complexity is an important indicator for assessing computational efficiency and resource consumption. Table VII reports the parameter count, per-sample GFLOPs, and test time of representative compared methods, together with the complexity and runtime of DRIDNet on all four datasets. This comparison offers a concise view of the efficiency–accuracy tradeoff among different HSI–LiDAR fusion models.

As shown in Table VII, the compared methods exhibit a broad spectrum of model sizes and computational demands. DRIDNet employs a moderate number of parameters, remaining substantially lighter than the most parameter-intensive baseline (FusAtNet) while being higher than several lightweight designs. A consistent trend is observed in terms

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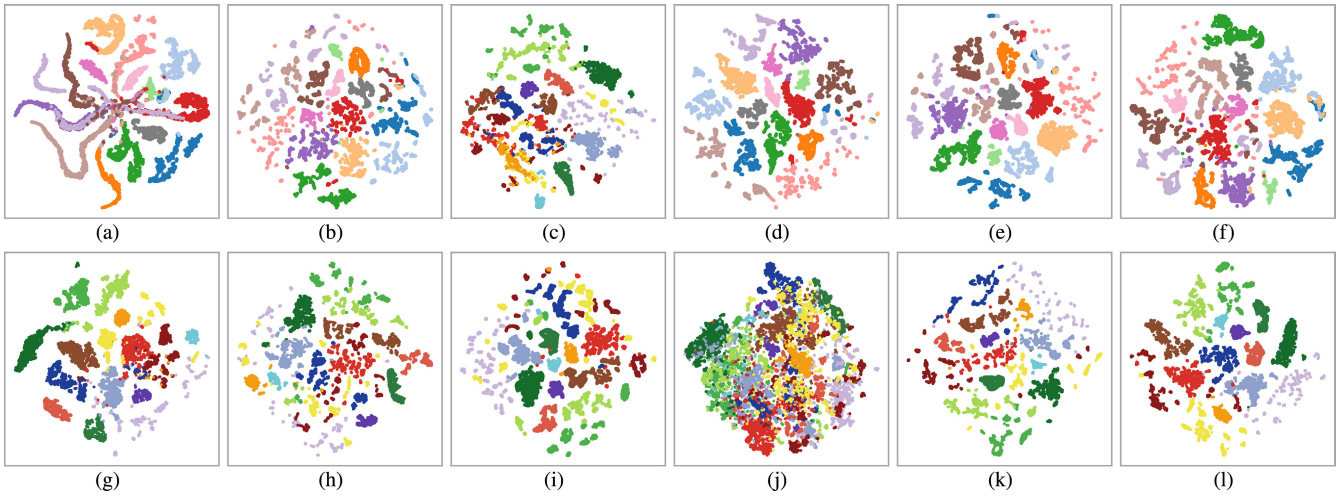


Fig. 12. t-SNE visualization of penultimate-layer features on the Houston2013 dataset. Panels are ordered as follows: (a) FusAtNet, (b) CALC, (c) HI2D2FNet, (d) MIViT, (e) FDNet, (f) M2FNet, (g) HyperDID, (h) S3F2Net, (i) DCMNet, (j) MCFNet, (k) E-Mamba, and (l) DRIDNet.

TABLE VII

MODEL COMPLEXITY AND RUNTIME COMPARISON. ALL COMPARED METHODS ARE EVALUATED ON HOUSTON2013, WHILE DRIDNET IS ADDITIONALLY REPORTED ON HOUSTON2018, MUUFL, AND TRENTO

Method	Params (M)	GFLOPs (per sample)	Test time (s)	Method	Params (M)	GFLOPs (per sample)	Test time (s)
FusAtNet	38.1092	6.9541	31.510	HyperDID	0.8460	0.0141	0.256
CALC	0.2841	0.0287	1.300	S3F2Net	0.3126	0.1788	8.102
HI2D2FNet	4.0551	0.1986	56.670	DCMNet	2.8399	0.0369	1.672
MIViT	2.6535	0.5882	6.650	MCFNet	0.2699	0.0393	1.781
FDNet	0.3845	0.0116	0.526	E-Mamba	1.7520	0.2162	2.783
M2FNet	0.2821	0.0916	4.150				
DRIDNet (Houston2013)	2.4374	0.0326	4.826	DRIDNet (Houston2018)	1.9522	0.3135	768.733
DRIDNet (MUUFL)	2.0605	0.1125	22.184	DRIDNet (Trento)	2.0375	0.0814	22.673

of GFLOPs: the proposed method requires more computation than the most compact baselines, but it is markedly lower than the heaviest model and thus stays within a practical computational budget. These results indicate that the performance gains are achieved without relying on excessive parameter stacking or prohibitively expensive computation.

In terms of inference efficiency, DRIDNet achieves a competitive test-time cost. Although it is not the fastest model, its test time is significantly reduced compared with the slowest baselines and remains within the mid-range of the compared methods. Together with the accuracy improvements reported in the previous subsection, the above results demonstrate that the proposed framework provides a favorable balance between recognition performance and computational complexity across the evaluated datasets.

H. Ablation Experiments

To provide a more complete understanding of the proposed framework, the ablation study is conducted from two perspectives: 1) the contribution of key modules in DRIDNet and 2) the effect of HSI and LiDAR input configurations. The first group of experiments verifies whether the performance gains arise from the four core modules, namely, CIIDM, SDDP, DPFEM, and CTRM. The second group evaluates whether the joint use of HSI and LiDAR is necessary for achieving the final performance.

1) Effect of Key Modules: To examine the contribution of each core component more clearly, we construct four ablated variants of DRIDNet, namely, w/o CIIDM, w/o SDDP, w/o DPFEM, and w/o CTRM. For each variant, the removed module is replaced by a dimension-preserving alternative, so that the downstream data flow and training protocol remain comparable to those of the full model. The results are summarized in Table VIII. The full DRIDNet consistently achieves the best OA, AA, and κ on all four datasets, demonstrating the effectiveness of the collaboration among intrinsic decomposition, spectral prior regularization, multiscale feature encoding, and cross-modal transport routing.

a) Ablation of CIIDM: For w/o CIIDM, the CIIDM is replaced by an identity HSI front-end. In this variant, no reflectance or shading component is estimated, and the observed HSI radiance is directly used as the effective HSI representation before PCA and feature encoding. Since no reflectance output is produced, the CIIDM-related losses and the SDDP prior are disabled, while DPFEM and CTRM are kept unchanged. As shown in Table VIII, this variant causes the largest OA degradation on all datasets, with drops of 1.95%, 1.52%, 2.36%, and 0.56% on Houston2013, Houston2018, MUUFL, and Trento, respectively. This confirms that CIIDM is critical for reducing shadow-induced spectral distortion and improving the robustness of multimodal learning.

b) Ablation of SDDP: For w/o SDDP, CIIDM is retained, but the learned SDDP is replaced with a handcrafted second-order spectral smoothness prior. Specifically, the diffusion-based prior loss is removed, and a second-order finite-difference regularization is imposed on each recovered reflectance spectrum to penalize abrupt spectral curvature. In this setting, the recovered reflectance is still constrained by the data fidelity loss, the DSM-guided smoothness loss, and a conventional spectral smoothness prior, but it no

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TABLE VIII
ABLATION ANALYSIS OF KEY MODULES IN DRIDNET

Cases	Houston2013			Houston2018			MUUFL			Trento		
	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
w/o CIIDM	92.36	92.91	91.71	92.54	95.24	90.79	88.07	89.72	84.93	98.76	97.87	98.49
w/o SDDP	93.14	93.68	92.62	93.48	96.05	91.75	88.94	90.66	85.86	98.98	98.16	98.77
w/o DPFEM	93.29	93.84	92.79	93.03	95.84	91.39	89.16	91.05	86.14	99.01	98.24	98.80
w/o CTRM	93.51	94.08	93.03	93.31	95.93	91.69	89.36	91.28	86.41	99.07	98.29	98.88
Full DRIDNet	**94.31**	**94.92**	**93.83**	**94.06**	**96.66**	**92.32**	**90.43**	**92.44**	**87.48**	**99.32**	**98.64**	**99.08**

TABLE IX
ABLATION ANALYSIS OF HSI AND LIDAR INPUT CONFIGURATIONS

Cases	Houston2013			Houston2018			MUUFL			Trento		
	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Only HSI	92.81	93.17	92.22	92.47	95.76	90.89	88.61	90.84	86.03	98.88	98.41	98.73
Only LiDAR	89.64	90.21	88.31	91.18	93.46	88.73	86.39	88.27	82.96	98.37	98.29	98.08
HSI+LiDAR	**94.31**	**94.92**	**93.83**	**94.06**	**96.66**	**92.32**	**90.43**	**92.44**	**87.48**	**99.32**	**98.64**	**99.08**

longer benefits from the learned diffusion-based reflectance manifold. Compared with the full model, w/o SDDP leads to OA drops of 1.17%, 0.58%, 1.49%, and 0.34% on the four datasets. This indicates that although second-order spectral smoothness can provide basic local regularization, it is less effective than SDDP in modeling global interband correlations, material-dependent absorption patterns, and physically plausible spectral distributions, thereby confirming the contribution of the learned diffusion prior.

c) Ablation of DPFEM: For w/o DPFEM, the DPFEM is replaced by a plain single-scale dual-stream encoder. Specifically, the HSI branch uses a standard 3-D convolutional block, and the LiDAR branch uses a standard 2-D convolutional block. Their outputs are projected to the same channel dimension as in the full model and then rearranged into tokens for fusion. This replacement keeps the dual-stream HSI–LiDAR structure but removes multiscale pyramid encoding and structured context aggregation. The OA drops by 0.90%, 1.05%, 1.28%, and 0.32% on the four datasets, confirming that DPFEM improves modality-specific representation learning by capturing hierarchical spectral–spatial and geometric cues.

d) Ablation of CTRM: For w/o CTRM, the CTRM is replaced by a simple concatenation-based fusion head. The top-level HSI and LiDAR features are concatenated along the channel dimension and then fed into global pooling and a linear classifier. This removes bidirectional cross-modal attention, Sinkhorn-based transport routing, and expert-wise adaptive aggregation, while preserving the final classification interface. The OA decreases by 0.68%, 0.78%, 1.05%, and 0.26% on the four datasets. These consistent drops indicate that simple concatenation is less effective than CTRM in aligning HSI and LiDAR features and adaptively allocating modality contributions across different regions and land-cover categories.

2) Effect of Modality Inputs: To examine the role of modality configurations, three input settings are evaluated on all datasets, namely, Only HSI, Only LiDAR, and the joint HSI + LiDAR input. As reported in Table IX, the joint HSI + LiDAR configuration consistently achieves the best performance in terms of OA, AA, and κ, whereas using a single modality leads to a clear degradation.

This comparison verifies the complementarity between HSI and LiDAR. HSI provides dense spectral signatures that are essential for material discrimination, whereas LiDAR contributes elevation and structural cues that are particularly helpful for separating classes with similar spectra but different geometric characteristics. Consequently, the Only HSI setting tends to under-represent height variation and structural boundaries, while the Only LiDAR setting loses fine-grained spectral information that is critical for distinguishing vegetation, roads, pavements, and mixed surfaces.

Compared with Only HSI, the joint HSI + LiDAR setting improves OA by 1.50, 1.59, 1.82, and 0.44 percentage points on Houston2013, Houston2018, MUUFL, and Trento, respectively. Compared with Only LiDAR, the gains are even larger. These results indicate that the proposed framework benefits substantially from jointly exploiting spectral and geometric information, and that the final performance cannot be achieved by either modality alone.

V. CONCLUSION

In this article, DRIDNet is proposed for shadow-robust joint classification of HSI and LiDAR data. Specifically, a CIIDM is designed to decompose the observed radiance into reflectance and a shading-related environmental component, and an SDDP is introduced to regularize the recovered reflectance toward a physically plausible distribution. In addition, a dual-stream pyramid encoder and a CTRM are developed to capture multiscale spectral–spatial and geometric cues and to achieve adaptive multimodal fusion. Experiments on four public benchmarks demonstrate that DRIDNet achieves OAs of 94.31%, 94.06%, 90.43%, and 99.32% on Houston2013, Houston2018, MUUFL, and Trento, respectively. Moreover, removing SDDP reduces OA by 1.17, 0.58, 1.49, and 0.34 percentage points on the four datasets, and the joint HSI–LiDAR input improves OA by 1.50, 1.59, 1.82, and 0.44 percentage points over the HSI-only setting, respectively, which further verifies the effectiveness of the proposed spectral

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regularization and multimodal fusion design. On the full shadow region of Houston2013, DRIDNet also improves the strongest baseline by 0.65 percentage points in OA, demonstrating better robustness under shadow-induced spectral distortion.

In future work, considering that DRIDNet relies on well-registered HSI–LiDAR inputs and sensor-band spectral priors, we will further investigate robustness to imperfect multimodal registration and cross-sensor spectral adaptation when extending the framework to broader remote sensing scenarios, including synthetic aperture radar, high-resolution optical imagery, and thermal infrared data. We will also explore more efficient implementations and self-supervised pretraining to further improve generalization under shadowed and low-label conditions.

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